Analytics 2: Advanced Predictive Techniques

NTPC Limited – A Case Study

Executive Summary

This report seeks to understand the common trends of electricity theft in India and recommend potential ways to detect such fraudulent usage.

A recent study reveals that nearly 20% of the total electricity generated in India is lost due to theft and fraud (Sharma et al., 2016). Electricity theft can take various forms, including tampering with electric meters, illegally tapping into power lines, and reconnecting disconnected meters. This issue is significant as these actions not only lead to financial losses but also pose serious safety risks, such as electric shocks, fires, and explosions.

Among all utility companies in India, NTPC (National Thermal Power Corporation) has been one of them who has been suffering from unprecedented underwriting loss due to significant electricity theft. Underwriting loss occurs when the company is unable to recover the shortfall in revenue due to fraudulent activities. This stolen electricity ultimately gets passed on to honest customers through higher tariffs. Therefore, this prompts NTPC to find alternative approaches that could aid in identifying instances of fraud, ultimately minimizing the risk of potential losses.

The dataset utilized for constructing our models was sourced from Kaggle, resembling NTPC's dataset in terms of information and variables for analysis. We conducted data cleaning in R and explored the data to gain insights and pinpoint key variables associated with fraudulent cases. Our analysis involved employing three models: Classification and Regression Trees (CART), Random Forest, and Neural Network.

In constructing our models, we utilized both unbalanced and balanced datasets. Upon analysis, we determined that the unbalanced CART model outperforms the balanced CART model in terms of accuracy and variable importance. Whereas for the Random Forest model, we found that the balanced Random Forest model outperforms its unbalanced counterpart in variable importance, accuracy, and error rates. When comparing the CART, Random Forest and Neural Network models, we recommend prioritizing Random Forest due to its highest accuracy rates.

Subsequently, we provide various practical recommendations for NTPC, including both technical enhancements and operational enhancements such as collaborative ventures and stricter policy implementations. Such actions can assist NTPC to cut losses from electricity theft and come up with better plans to help the company combat such issues. For a well-rounded analysis, we acknowledge the constraints encountered in the use of predictive techniques and models from a business perspective.

Lastly, we assess the efficacy of our suggested solutions and their potential to enhance NTPC's fraud detection framework. We evaluate the effectiveness of our solutions through examining changes in NTPC's underwriting amounts, specifically focusing on loss in profits attributed to electricity theft and the number of fraudulent cases detected.

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1. Introduction

1.1 NTPC - Leading Energy Provider in India

NTPC Limited, formerly National Thermal Power Corporation, is a power company in India. NTPC's core function is the generation and distribution of electricity to state electricity boards in India. Currently, NTPC is the largest power company in India with an electric power generating capacity of 71,594MW, which contributes to an estimated 25% of the total power generation in India (Appendix A).

The company currently owns and operates 26 coal-based stations, seven gas-based stations, one hydro station, one small hydro station, 11 solar PV stations, and one wind-based station. In addition, they help operate 25 Joint Venture stations (nine coal-based, four gas-based, eight hydro-powered, one small hydro, two wind-powered, and one Solar PV).

2. Business Problem

2.1 Non-Technical Loss – Meter Tampering

Electrical energy meter tampering refers to any form of alteration, manipulation or modification done to an electrical energy meter, which results in inaccurate meter readings (Shenzhen Clou, 2023). This is usually done to reduce or avoid the amount of electricity that the consumer is charged for. This action is illegal and results in non-technical loss, which is a major problem in the energy industry.

In India, electricity theft, often through meter tampering, is significantly driven by the lack of precise energy audit and accounting for consumers. The traditional method of collecting meter readings, which relies on personnel visiting each customer to record consumption data, presents several challenges. This manual process is not only time-consuming and labour-intensive but also prone to errors due to human involvement, making it an inefficient aspect of billing and monitoring electricity usage.

For a considerable period, power companies have grappled with maintaining direct engagement with their consumers. Till this day, the Indian power sector still faces numerous challenges, primarily due to electricity theft and network inefficiencies. A major concern has been the outdated billing system, characterized by its inaccuracy, inflexibility, and unreliability. Consequently, there is an ongoing push towards automating the billing system, a task pursued with continuous effort. Although advancements have been made in meter reading accuracy, the billing and payment processes still largely follow outdated practices, necessitating further improvement. Currently, the process involves a representative visiting the consumer's premises to note the meter reading and prepare a report, which then determines the payment amount. However, this lengthy procedure is prone to errors. This paper discusses the issues arising from the lack of effective communication systems and explores potential benefits and solutions to these challenges.

2.2 Consequences of Meter Tampering on different stakeholders

2.2.1 General community

Ultimately, the impact of meter tampering will be felt by all consumers, as the revenue losses incurred by these providers as well as the cost of additional measures means that they are forced to pass these costs to consumers by driving up electricity rates in the long run to cover their losses and remain profitable. In the United Kingdom, over £400 million worth of energy is stolen annually, with projections suggesting an extra cost of between £20 and £30 for each “honest customer’s bill” (LeicestershireLive, 2022).

2.2.2 Utility companies

It is estimated that utility companies worldwide lose more than $25 billion every year due to electricity theft, of which utility companies in India make up nearly 20% ($4.5 billion) of this amount. For NTPC alone, it is estimated that the company suffered an Aggregate Technical and Commercial (AT&C) loss of 15.41% in 2023. This translates to about $3 billion worth of lost revenue (Mercom, 2023). Besides NTPC, other power companies in India also suffer huge loses in revenue due to widespread electricity theft. Some of these companies may be even operating at a net or even a loss. Should this trend worsen, power companies will be forced out of business individually.

Meter tampering also results in more costs incurred by utility companies such as NTPC towards preventative and corrective efforts being put in to combat electricity theft. An example of this would be NTPC’s aim to implement smart meters in all consumer premises by 2024. These smart meters are estimated to cost approximately $73 USD per unit with the whole project being expected to cost nearly $17 billion, which is more than 1 years' worth of NTPC’s revenue.

2.2.3 Government

Meter tampering also presents an additional issue of safety, as meter tampering can lead to overloaded circuits, electrical fires and electrocution hazards (Günther, 2023). This is further exacerbated by the fact that reduced profits means that there is a greater incentive for these companies to cut cost, which could lead to lower quality equipment and lead to more power outages and decreased reliability. Ultimately, meter tampering causes not only economic losses to the electricity providers involved, but also poses a serious safety hazard to consumers alike.

3. Opportunity and Objective

Our team aimed to implement machine learning to analyze past data on electricity theft so NTPC can forecast the probability of theft along its power lines and detect fraudulent customers. Identifying and utilizing patterns in energy consumption allows us to detect unusual behavior. Hence, integrating live electricity consumption data into this predictive model presents a cost-effective alternative to 2 main costs of NTPC’s efforts to combat electricity theft.

1) We reduce expenses by minimizing the need for inspectors to physically visit every household to verify the functionality of electricity meters.
2) Our solution acts as a much cheaper alternative to widespread meter upgrades. This method not only diminishes the necessity for extensive manual inspections but also improves the speed and precision in detecting theft incidents.

Moreover, this innovative approach provides NTPC with an opportunity to lead the transition towards using machine learning and analytics to combat electricity theft and meter tampering. Adopting data-driven strategies positions NTPC as an industry innovator, potentially opening new revenue channels. For example, NTPC could extend these prediction models as services to other utilities grappling with similar issues, creating a novel business vertical. Additionally, securing IP rights for these solutions could enable NTPC to license the technology to others, further diversifying its income sources and cementing its stature as a forward-thinking pioneer in the energy domain. This strategy not only aims at reducing current losses but also envisions a future where NTPC sets industry benchmarks in leveraging technology to address challenges related to electricity theft and meter tampering. Finally, the findings can assist NTPC in strategically prioritizing the installation of smart meters in regions where more fraud is identified.

3.1 Project Approach

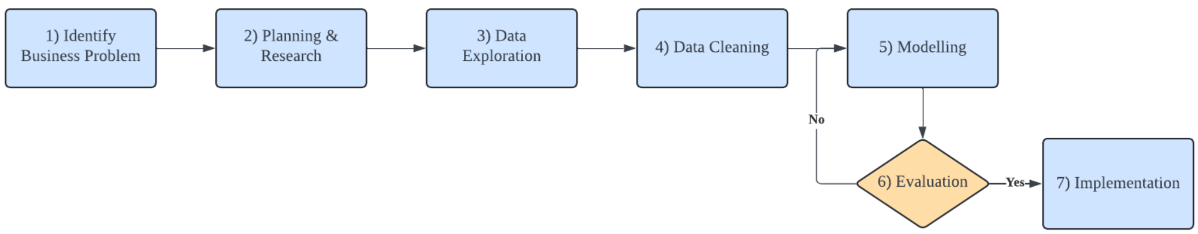


Figure 1: Flow chart for project

Stage 1 involves identifying the business problem, which sets the foundation for subsequent stages as mentioned in section 1 and 2. Once the problem is defined, Stage 2 kicks in, emphasizing thorough research and planning as represented in section 4. This phase is crucial as it delves into both technical and non-technical existing solutions while identifying potential business opportunities. Brainstorming sessions are conducted to devise strategies before moving forward to Stage 3.

In Stage 3, data exploration takes precedence. This involves delving deep into the data to uncover relationships and key insights. Visualizations are employed to make these findings more tangible and actionable, serving as valuable inputs for crafting solutions tailored to NTPC's needs.

Moving on to Stage 4, the focus shifts towards data processing. Here, the data is cleaned, and relevant features are selected in preparation for building predictive models. The goal is to ensure that the models are trained on accurate and meaningful data, laying the groundwork for robust predictions.

Stage 5 marks the curation of various predictive models aimed at identifying potential instances of meter tampering by NTPC's clients. Models such as CART, Neural Network, and Random Forest are employed, each offering unique advantages. The diversity of models allows for comprehensive evaluation, ensuring that the chosen model delivers optimal accuracy. Additionally, simpler models like CART contribute to explainability, helping to demystify complex predictions.

Stage 6, the evaluation phase, utilises 3 metrics to consider the accuracy of each model and determine which is the best.

1) **Best Overall Accuracy:** This metric identifies the overall accuracy rate of the models in predicting meter tampering cases by focusing on the proportion of accurate predictions. $Overall\ Accuracy = \frac{(True\ Positive + True\ Negative)}{Total}$
2) **Highest True Positive Rate:** This metric identifies how many actual cases were identified correctly out of all the meter tampering cases. High accuracy is preferred or else fraudulent

NTPC clients will go undetected.

$True\ Positive\ Rate\ (Recall) = \frac{True\ Positive}{(True\ Positive + False\ Negative)}$

3) **Highest Precision Rate:** The metric measures the proportion of correct predictions in relation to all the predictions made. This is crucial as a low percentage would indicate that there is inefficient use of manpower, where frontline workers visit substantially more households than required, incurring greater cost for NTPC.

   $Positive\ Predictive\ Value\ (Precision) = \frac{True\ Positive}{(False\ Positive + True\ Positive)}$

Lastly, once the preferred model has been selected, it will be used for implementation through a Power Bi dashboard as seen in section 6.

4. Current Solutions

This section aims to identify the advantages and disadvantages of the two most relevant current solutions, forming as a comparison to the proposed solution.

4.1 Governmental Approach - Smart Meters

The Smart Meter is an advanced energy meter as well that can identify energy consumption more accurately compared to the traditional ones. This meter is also able to link securely to the local utility for monitoring and billing purposes. It is also claimed to be particularly hard to tamper and hack and thus provide a source of safety for the companies (Warudkar et. al., 2014). Furthermore, there has been a mandate by the Indian Ministry of Power for the installation of Smart meters throughout the entire country for every household, illustrating its relevance and significance (Jones, 2024).

Advantages of Smart Meters

1) Identify energy consumption more accurately compared to the traditional meters

2) Reducing manpower as meters are linked to the local utility for monitoring and billing (Clou, 2023).

Disadvantages of Smart Meters

1) Despite claims that they are difficult to tamper and hack, there are still several cases of theft through bypassing their systems. It is claimed that smart meters could be hacked easily as mimicking communication devices could learn how to communicate with the meter and access it through malware once any meter has been compromised (Latief, Y., 2023). Hence, defeating its purpose.

2) These technical solutions also incur high costs for the company, and this raised the question of "with more than 90% of consumers spending less than INR 700/month today, do we really need a smart meter?" (Khaitan, Y., 2024).

3) The RDSS, a government-led initiative, plans to introduce smart meters across all consumer areas as a measure against electricity theft. However, this strategy comes with a significant expense, with an estimated cost of 17 billion dollars solely for the installation of the meters (Verma, P., 2023).

4.2 NTPC Internal Approach

Currently, NTPC uses a 2-pronged approach in detecting fraud claims. In the 1st layer, frontline staff are equipped with decision rules based on common characteristics of fraud. In the 2nd layer, suspicious cases are flagged and escalated to the Vigilance Department of NTPC. When cases are flagged, the Vigilance Department first analyses electricity usage data to determine if it is fraud. If a case is highly likely fraud, investigations commence to attain evidence.

Advantages of 2-Pronged Approach
1) **Efficient frontline detection:** Frontline staff can readily identify potential fraud cases during their routine meter collection, enhancing the efficacy of meter tampering detection to ensure timely investigation and action
2) **Cross-validation:** The 2-pronged approach allows for cross-validation in reducing false positives flagged by the frontline staff to ensure that only legitimate fraud cases proceed to evidence collection for investigation

Disadvantages of 2-Pronged Approach
1) **Heavy dependence on staffs' judgement:** prone to human error & lapses, leading to false positive cases which consume investigation time & resources that could be allocated to genuine fraudulent activity.
2) **Time-consuming** and resource intensive framework to identify fraud based on a complex set of decision rules. With limited staff, resources, and a narrow profit margin, it is impossible to do in-depth analysis on each consumer without jeopardizing the company's profitability.

These limitations coupled with the increasing complexity of fraud schemes have caused fraud cases to spike, undetected by existing anti-fraud measures.

5. Data Exploration

5.1 Dataset

The dataset "Fraud Detection in Electricity and Gas Consumption" is obtained from Kaggle. The dataset includes over 200,000 records with more than 20 attributes (Appendix B).

5.1.1 Relevance of Dataset to NTPC

NTPC does not share their dataset for privacy reasons, hence the dataset used for model construction obtained from Kaggle, is one which is like that of NTPC's dataset in terms of variables and information utilized for analysis. Hence, we feel that the dataset is extensive enough to conduct a thorough analysis. The result variable (target) is either 0 or 1. This dataset will enable us to do thorough exploration and model building to uncover useful insights. Furthermore, there is a vast amount of data available that may be used to understand issues and gain insights related to fraud cases in the Utilities industry.

5.2 Exploratory Data Analysis

We wanted to obtain a more comprehensive and in-depth understanding of the dataset, the correlations among various factors, and the distributions and trends among important variables.

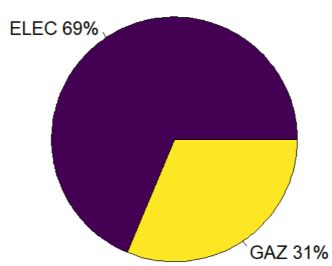


Figure 2: Electricity and Gas distribution pie chart

To start off, the pie chart shows the proportion of counter type (electricity to gas) amongst all consumers. The data shows that 69% of dataset are for electricity and 31% are for gas.

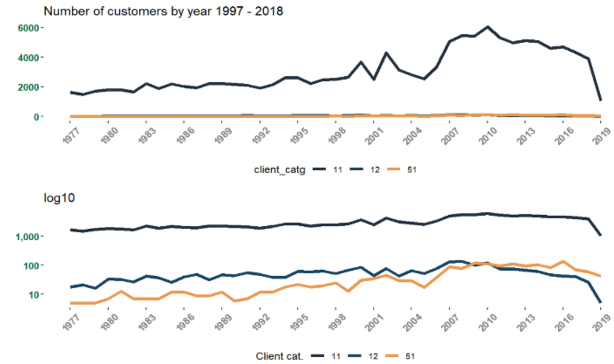


Figure 3: Graph of number of customers by year

From Fig 3, the number of customers increased from 1977 to 2019, with the largest increase of customers in category 11. This category is likely to be made up of residential customers, as this is the largest category of customers for STEG. From 2006 onwards, there has been a more noticeable increase in the number of customers, and we noticed the same for the client categories 12 and 51.

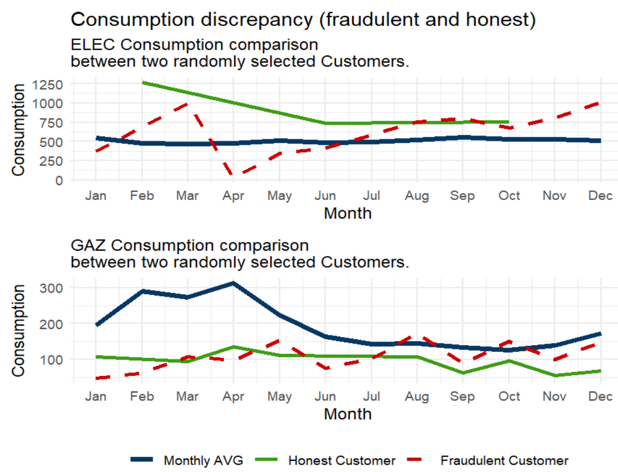


Figure 4: Yearly Consumption Pattern

The plot displays the average consumption level 1 for two randomly selected customers, one fraudulent and one honest, over a range of dates in the year 2005. We observe that the green line for the honest customer is relatively stable with only slight fluctuations over time, whereas the red line for the fraudulent customer is much more erratic and unpredictable, with frequent spikes and dips in consumption. This suggests that the fraudulent customer is likely tampering with their meter to underreport their actual consumption levels. Overall, the plot provides a clear visual comparison of the consumption patterns of the two customers and highlights the stark differences between them.

Figure 5: Counts of honest non-fraud and fraud cases

The distribution of honest(0) and fraud(1) cases is clearly imbalanced, with far more non-fraud than fraud cases. This can be a problem for machine learning models, as they can be biased towards the majority class. Hence, undersampling will be conducted during modelling, which will be explained later.

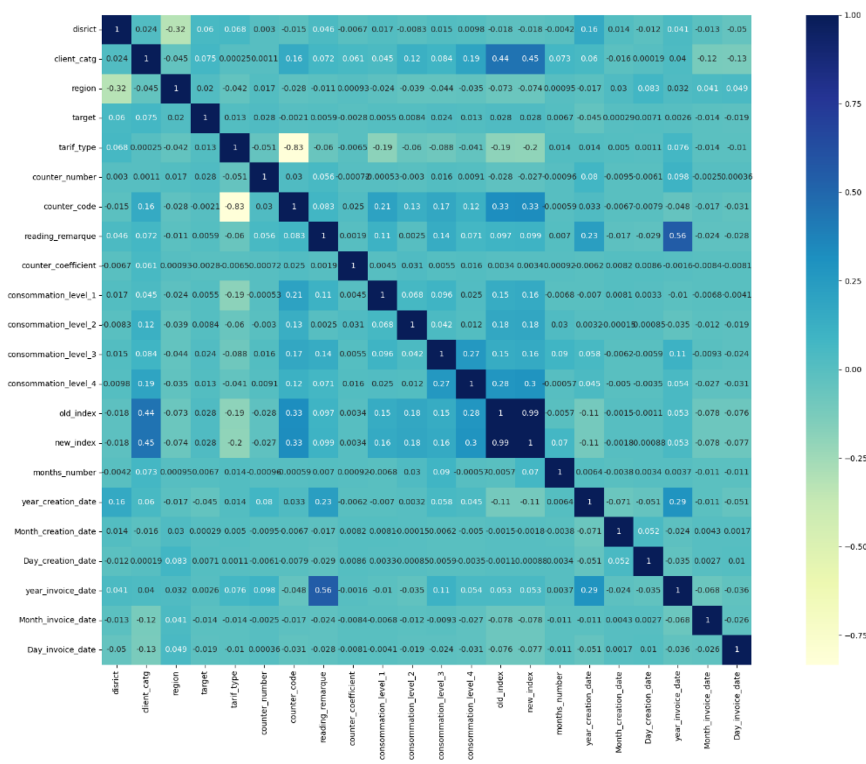


Figure 6: Correlation Matrix

The consumption levels 1 to 4 were all positively correlated with each other, with a correlation range between 0.012 to 0.27. Additionally, "old_index" and "new_index" positively correlated, with a high correlation value of 0.99. The "target" variable showed the highest positive correlation with "client_catg" and "district", with values of 0.075 and 0.06 respectively.

Additionally, a negative correlation has been noticed between some attributes including "counter_code" and "tariff_type", which has the largest negative correlation detected with a value of -0.83.

5.3 Data Preparation

Data preparation is an important and the foundation for the subsequent model building. This helps to ensure results are accurate and reliable. There would be multiple processes for the Data preprocessing in this Data preparation stage. This includes data cleaning, data transformation, feature engineering and feature selection. This stage primarily focuses on 2 of our data sources, invoice_train & client_train.

5.3.1 Data Preprocessing

5.3.1.1 Data Cleaning

Firstly, the dataset underwent column exploration and transformation to categorical variable types, as depicted in Appendix C. Subsequently, an examination for missing and null values was conducted, revealing 13 NA values within the "counter_statue" column. It was decided to replace these NA values with the mode of the column, which was found to be "0". Following this, data integrity was further ensured by scrutinizing for erroneous values utilizing both the summary() and unique() functions. This process was particularly instrumental in identifying discrepancies in critical variables such as electricity consumption. Duplicates were also found and removed accordingly (Appendix D).

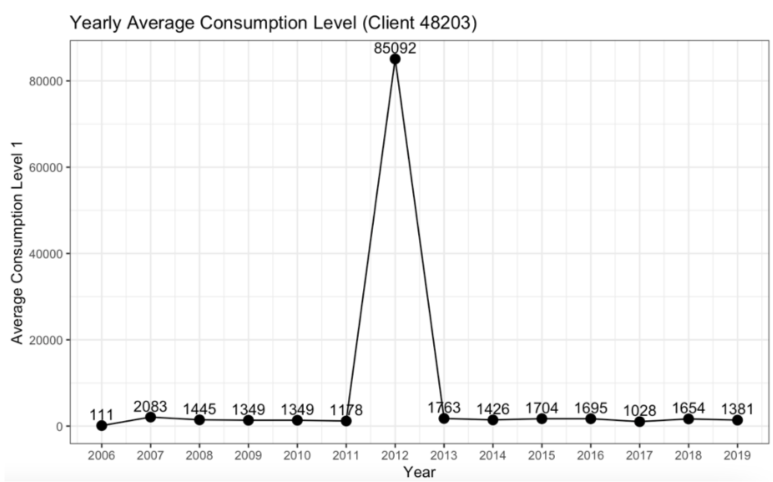


Figure 7: Plot of Consumption level 1 outlier customer; to take average of year before and after outlier

Lastly, outliers were also identified in the column's consumption levels 1 to 4 (Appendix E). To replace the values for the outliers, the consumption value was determined based on the average of the previous and following year's consumption; consumption level 1 example seen in Figure 7. This approach is justified where Singapore was taken as inspiration. It adopts a bi-monthly meter readings are taken and replaced by average value for months that are not being read (EMA, n.d.). Similarly, we apply the same concept for the outliers that may have been potentially typed wrongly to the average of the previous year and after year of outlier average energy consumption.

5.3.1.2 Feature Engineering

Feature engineering is the process of selecting and transforming variables for the models. This will be able to enhance the predictive capability of the models and reveal any hidden behaviour and patterns.

To prepare the data for modelling, we need to combine the data sources to a single dataset base on their primary key of client_id. However, client_train contains details of each distinct customer but invoice_train includes several invoice details for every customer (Figure 8). Therefore, there are multiple ways to group these attributes for each customer in the invoice_train dataset. This can be done coherently with creating new features for these variables to collect more information and generate more insights.

	disrict	client_id	client_catg
1	60	train_Client_0	11
2	69	train_Client_1	11
3	62	train_Client_10	11

	client_id	invoice_date	tarif_type
1	train_Client_0	2014-03-24	11
2	train_Client_0	2013-03-29	11
3	train_Client_0	2015-03-23	11

Figure 8: Sample of client_train (left) & invoice_train (right)

This is done by utilising statistical features like sum, diff, average, variance, standard deviation, median, mode and range of each non-categorical attribute in the invoice_train data source. The attributes include the consumption levels 1-4, old & new index, invoice_date, counter coefficient, counter status and reading remarks. These new attributes are then added to the dataset. These attributes are merged under a distinct client ID. However, counter_types result in another row sharing the same client ID. Hence, the counter_type has been merged with these attributes (Figure 9).

	client_id	counter_type	avg_consom_1
1	train_Client_0	ELEC	352.4000000
2	train_Client_1	ELEC	557.5405405
3	train_Client_10	ELEC	798.6111111
4	train_Client_100	ELEC	1.2000000
5	train_Client_1000	ELEC	663.7142857
6	train_Client_10000	ELEC	422.0689655
7	train_Client_10000	GAZ	245.3157895

	client_id	avg_consom_1_ELEC	avg_consom_1_GAZ
1	train_Client_0	352.4000000	0.000000
2	train_Client_1	557.5405405	0.000000
3	train_Client_10	798.6111111	0.000000
4	train_Client_100	1.2000000	0.000000
5	train_Client_1000	663.7142857	0.000000
6	train_Client_10000	422.0689655	245.315789

Figure 9: Merging of counter_type with attributes

5.3.1.3 Data Transformation

After the full_data_model has been created after merging. It is essential for us to transform the data to the appropriate format for further data analysis.

5.3.1.4 Feature Selection

It is necessary to identify features that are useful for our analysis. There are 4 criteria we use to evaluate these features. First, their statistical value in terms of importance which we would be evaluating later in the models. Second, **industry requirements** on whether these features are being regarded or often tracked. Third, we identify any relationship based on our data exploration and lastly, the features practicality, i.e. whether they can be adjusted or collected in real-world scenarios.

After evaluating all these criteria, we deemed 3 attributes to not be of value to the target variable: “Client_ID”, “counter_number” and “counter_code”. These variables only provide information related to administrative bookkeeping reasons and do not provide analytical insights. These attributes were removed from the dataset do not seem to have a relationship with our target variable. We replaced “Client_ID” with “tariff_type” to identify each row of records. The rest of the attributes would be adjusted accordingly as per the information gained on the model algorithm.

5.3.1.5 Train Test Split & Downsampling
For the forthcoming model training, accuracy and related metrics will be assessed using a 70:30 train-test split. The initial dataset, comprising 127,927 "0" observations and 7,566 "1" observations, has undergone downsampling to achieve a balanced representation of "0" and "1" observations.

5.3.2 Final full_data_model
This leads to the final dataset of full_data_model with 204,982 rows and 137 columns (Appendix F).

6. Modelling

6.1 CART
The CART model employs a method of successive splitting of nodes in a decision tree to predict the outcome variable by dividing the data into increasingly homogeneous subsets based on the independent variables. Due to the imbalance of the dataset discovered during exploratory analysis, two CART models will be developed - one using the original unbalanced dataset and another with a balanced dataset - to predict the categorical target variable.

6.1.1 CART model with Unbalanced Data
Our comprehensive initial CART model, which incorporated all variables, yielded a total of 5104 splits. We then applied weakest link pruning by incorporating a complexity penalty (CP) into the cost calculation of the model to identify the most suitable pruning point (see Appendix G for details). This process resulted in the selection of the 2249th split as the optimal point for pruning, a decision made through computation. Utilizing the pruned version of the tree, we conducted an analysis of variable importance and discovered 20 significant variables that contribute to the prediction of the target variable. Within this model, "region" emerged as the most critical predictor (detailed in Appendix H). It is noted that using this pruned CART model on unbalanced data achieved a prediction accuracy of 94.7%.

Number of test data = 61470		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	1744	2129
	No (Not Meter Tampering)	1128	56469
Positive Predictive Value (Precision)		60.7%	
True Positive Rate (Recall)		45.0%	
Predictive Accuracy		94.7%	

6.1.2 CART model with Balanced Data
Another CART model is trained with the balanced data set. This results in a model grown to 2082 splits and eventually pruned at 1114 split based on the optimal CP calculated. This model has an accuracy of 77.12%.

Number of test data = 7710		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	3088	769

	No (Not Meter Tampering)	995	2858
Positive Predictive Value (Precision)		75.6%	
True Positive Rate (Recall)		80.0%	
Predictive Accuracy		77.12%	

Similar methods to determine the significant variables are conducted, and the balanced model produced 19 significant variables (Appendix I). These variables are used to train the balanced dataset again. It is interesting to find out that the balanced model trained with only 19 variables yields a similar result to the 137 variables. Only a minor decrease in accuracy from 77.12% to 75%.

6.1.3 Visualisation from Balanced CART Model

Plot for the first few splits were visualized from the balanced CART model. There are 3 splits which we can use to observe how the model classifies fraudulent electricity consumption.

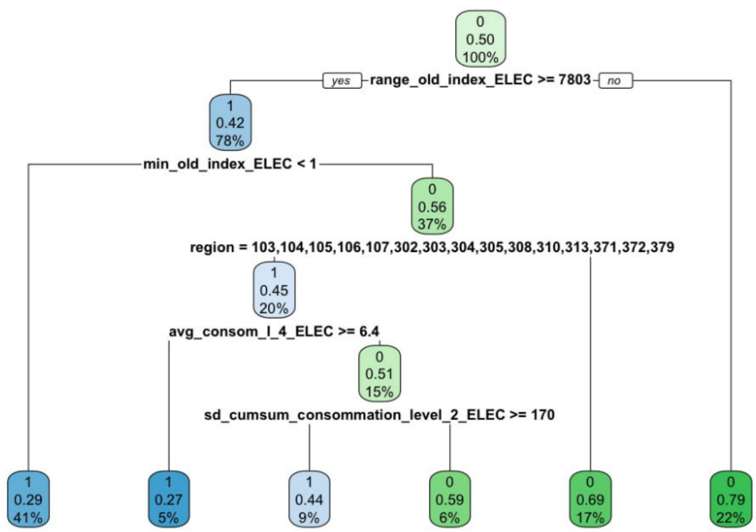


Figure 10: Visualisation of first few split of balance CART model

1. The case is predicted to be a fraud (1) with 29% of Y=1 and 41% of total observations if the “range_old_index_ELEC” is more than or equals to 7803 and if “min_old_index_ELEC” is less than 1.
2. The case is predicted to be a fraud (1) with 45% of Y=1 and 20% of total observations if “range_old_index_ELEC” is more than or equals to 7803, “min_old_index_ELEC” is more than 1 and "region” are 103,104,105,106,107,302,303,304,305,308,310,313,371,372,379.
3. The case is predicted to be a fraud (1) with 27% of Y=1 and 5% of total observations if “range_old_index_ELEC” is more than or equals to 7803, “min_old_index_ELEC” is more than 1, "region” are 103,104,105,106,107,302,303,304,305,308,310,313,371,372,379 and “avg_consom_l_4_ELEC” is more than or equal to 6.4.

Based on the few splits above, we can identify the root node of “range_old_index_ELEC”. Other variable such as “min_old_index_ELEC” and “region” are crucial in predicting electricity theft in the balanced CART model. If a customer has a high value of above 7803 for “range_old_index_ELEC”, they are most probably fraudulent. In terms of regions, if they are in regions 103, 104, 105, 106, 107, 302, 303, 304, 305, 308, 310, 313, 371, 372, 379 and have “min_old_index_ELEC” more than 1 and “range_old_index_ELEC” is more than or equals to 7803, they are probably fraudulent.

6.1.2 Evaluation of CART Models

The balanced CART model appears to perform more effectively overall, evidenced by enhanced precision and recall. Despite this, the model exhibits a slightly reduced predictive accuracy compared to its unbalanced counterpart. It is pertinent to mention that the application of a balanced dataset in classification models is generally advocated for, as it frequently yields models with superior accuracy, more equitable accuracy across classes, and more consistent detection rates (Amruthnath, 2020). The use of a balanced dataset is beneficial in counteracting the disproportionate influence that often accompanies unbalanced data.

Nevertheless, it should be acknowledged that an accuracy rate of 77.12% for the balanced CART model does not stand out as particularly high, suggesting the exploration of alternative modelling approaches may be prudent.

When comparing the 2 CART models, we observe that there are differences in variables selected using the variable importance analysis. Using a constant parameter of scaled variable importance more than or equal to 2, the balanced model can capture 3 additional important variables. This is crucial as this helps in making better classification in the model. These 3 variables which the unbalanced model failed to identify suggest how the balanced model might be a better option among these two.

The 3 additional variables are:

- “min_old_index_ELEC”
- “avg_cumsum_new_index_ELEC”
- “median_cumsum_old_index_ELEC”

6.2 Random Forest

Random Forest is an ensemble machine learning method employed for predicting either categorical or continuous variables. It combines bagging and Random Subset Feature (RSF) selection techniques. Bagging involves generating "B" bootstrap samples by randomly selecting with replacement from an original sample. Subsequently, "B" predictive models are built, each providing a prediction for the target variable Y. The final prediction is made by aggregating the predicted Y values from all "B" models. For categorical targets, the output is determined by majority voting across all trees, while for continuous variables, it relies on the mean prediction.

Within the "B" predictive models, CART is the most used amongst other models like logistic regression and linear regression. In addition to bagging, Random Forest incorporates RSF, where a subset of X variables is randomly chosen at each split in a CART model tree to identify the optimal split. RSF introduces controlled variability to reduce inter-tree correlation, preventing any single variable from dominating and improves model diversity without compromising on accuracy.

6.2.1 Random Forest Model with Unbalanced Data

We created a Random Forest Model using the unbalanced trainset which yielded the following results (Appendix J). OOB error rate is 2.71%. The plot for this RF model depicts this low OOB error rate, which is very close to 0, this means an almost perfect classification on OOB sample (Appendix L2). This may suggest overfitting even if it is less common in RF models since it is ensemble in nature.

Thereafter, the trained model was used to predict the test set. Overall accuracy of 97.3% was achieved with a low misclassification rate of 2.7%. However, there is low recall rate. This caused by an imbalance of the target variable in the trainset, causing the model to favour predicting the majority class (no fraud).

Number of test data = 61470		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	2278	1595
	No (Not Meter Tampering)	22	57575
Positive Predictive Value (Precision)		99.0%	
True Positive Rate (Recall)		58.5%	
Predictive Accuracy		97.3%	

6.2.2 Random Forest with balanced data

A new Random Forest Model was developed using the balanced trainset, and the outcomes are: OOB error rate increased to 19.13%. The plot for this RF model shows OOB error rate starts higher and decreases to a level that seems to stabilize around 15-20%. This indicates good performance, which may suggest a more balanced or realistic model performance (Appendix L4). Overall accuracy of 84.2% was achieved with a low misclassification rate of 15.8%. There is a lower overall accuracy rate but higher recall rate as compared to the unbalanced model due to under sampling (Appendix K)

Number of test data = 7710		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	3308	549
	No (Not Meter Tampering)	666	3187
Positive Predictive Value (Precision)		83.2%	
True Positive Rate (Recall)		85.7%	
Predictive Accuracy		84.2%	

This is to be expected since the new balanced data set has fewer observations overall, and a higher proportion of fraud cases (in comparison to the unbalanced model). As a result, there are more false positives and false negatives, and the overall accuracy is reduced.

The variables of this balanced random forest model were evaluated and were trimmed down to 19 variables based on their MeanDecreaseAccuracy. It was found out that the 19 variables have almost the same accuracy as the model with 137 variables, with a slight decrease in accuracy from 84.2% to only 82.2% (Appendix L). This allows the team to focus on these few variables in the solution section.

6.3.2 Evaluation of Random Forest Models

There are two differences between the balanced and unbalanced Random Forest models (Appendix N). First, there is a difference between the important variables identified. By observing the MeanDecreaseAccuracy, Variables like "count_invoice_date_GAZ" & "count_invoice_date_ELEC" were identified in the balanced model. These variables, however, are not captured in the unbalanced model.

There is a difference in accuracy & error rates. The unbalanced model had a higher accuracy rate of 97.3%, which is expected as it was trained with a bigger dataset and more observations. However, the FNR (41.94%) was high as the target variable was unbalanced. The majority class is often favored by the model during bagging and RSF due to its larger sample size, hence causing the FNR to be high.

On the other hand, the balanced model produced a lower accuracy rate of 84.2% as it had a smaller dataset with a lower number of observations. In the balanced model, we observe that FNR is low while FPR is high. This is due to random under sampling which removed more than half of the majority class data for the trainset to be balanced. The removal of 125,336 "0" observations meant a loss of important data that could have been used by the model to better predict "0". This caused a huge increase in FPR. On the other hand, FNR decreased due to a larger proportion of "1" cases, hence disallowing the model to favor the majority class through majority voting.

6.3.3 Business Evaluation: Random Forest

The recommended model for Random Forest is the balanced model, although it has a lower accuracy rate due to lower FNR. This is because the aim of the project is to detect fraudulent customers, hence a lower false negative is ideal. high false negative implies that numerous instances of fraud are undetected by the model, which can prove costly for the company over time. From a business perspective, investigating a false positive case is less costly than allowing undetected electricity theft to persist. Hence, while the balanced model may exhibit a higher FPR, this is deemed an acceptable trade-off.

6.4 Neural Network

6.4.1 Neural Network Model Optimisation

A neural network represents is an unsupervised machine learning model that enables computers to analyse data through a process modelled after the human brain. It operates through a network of interconnected nodes or neurons, arranged in layers like the neural architecture of the brain (Appendix M). This structure allows the system to adapt and refine its performance over time by learning from errors. Neural networks rely on training data to learn and improve their accuracy over time.

This is done by firstly scaling the trainset because it helps prevent biased updates of the model weights. When features in the dataset have varying scales, those with larger scales may disproportionately influence the weight updates, causing the network to focus more on learning from these features. By scaling, we ensure that each feature has an equal opportunity to contribute to the loss function. This uniform contribution allows the neural network to learn more efficiently and effectively from the entire dataset. This is done via the codes shown below

```r
# --- Data Preparation ---
# Convert factor variables to dummy variables
data_transformed <- data.frame(model.matrix(~ . - 1, data = trainset.bal))

# Normalize the data
preproc <- preProcess(data_transformed, method = c("center", "scale"))
data_normalized <- predict(preproc, data_transformed)

# --- Dataset Splitting ---
# Splitting the dataset (assuming 'target' is the last column)
target_variable <- data_normalized[, ncol(data_normalized)]
features <- data_normalized[, -ncol(data_normalized)]
```

Figure 11: Data preparation for neural network

To measure the error between computed outputs and the desired target outputs of the training data, we used cross entropy which measures the difference between two probability distributions for a given random variable or set of events.

6.4.2 Neural Network with Balanced data

The resulting neural network (Appendix N) consists of 2 hidden layers. Input weights of 3.96 and 7.13 are used for the first and second neuron of the first hidden layer. Also, input weights of 3.21 and -3.21 are used for the first and second neuron of the first hidden layer respectively.
The targets variable in the dataset was balanced using under sampling to create a 1:1 ratio to address the significant false positive rate.

We randomly divide the dataset into train and test set ratios of 70–30 with set.seed(1). The trainset includes 137 variables, including the target variable. Thereafter, the train model was used to predict the test set. Overall accuracy of 65.7% was achieved with a misclassification rate of 34.3%. From the results, we see that the model is better at predicting positive cases (meter tampering cases) than negative cases. This may be due to the down sampling of negative cases to ensure a 1:1 ratio which resulted in the potential loss of information. Also, model complexity might be a factor as well as 2 hidden layers may result in the model being not complex enough (increasing hidden layers increases processing time significantly given the sheer volume of data).

Number of test data = 18076		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	6713	2325
	No (Not Meter Tampering)	3874	5164
Positive Predictive Value (Precision)		74.2%	
True Positive Rate (Recall)		63.4%	
Predictive Accuracy		65.7%	

6.4.3Neural Network Model with Unbalanced Data

The dataset was randomly split into a train & test set ratio of 70-30 using set.seed(1). The imbalance in the target variable was left unbalanced.

We created a Neural Network model using the unbalanced trainset which yielded the following results. Thereafter, the train model was used to predict the test set. Overall accuracy of 76.2% was achieved with a low misclassification rate of 23.8%. However, the FNR was 81.2%. The high FNR is caused by an imbalance of the target variable in the trainset causing the model to favour predicting the majority class (no fraud).

Number of test data = 61470		Predicted	
		Yes (Meter Tampering)	No (Not Meter Tampering)
Actual	Yes (Meter Tampering)	3257	616
	No (Not Meter Tampering)	14033	43564
Positive Predictive Value (Precision)		83.1%	
True Positive Rate (Recall)		18.8%	
Predictive Accuracy		76.2%	

6.4.4 Business insights of Neural Network model

The low accuracy of the neural network model may be due to a few reasons. Firstly, due to the large number of input variables, the model might be adjusting for too much noise where they value the 'wrong' features. Also, the current network architecture that we have may not suit the dataset very well although adjusting it takes a very long time (increasing stepmax, increasing number of hidden layers and neurons). While for now, it may seem that CART performs better in predictive positive cases, it should be noted that an advantage of the neural network model comes in its adaptability in learning. With more volumes of data given and more time, it allows for the model to learn more and increase its predictive accuracy.

6.5 Evaluation & Selection of Models

Evaluation of Models			
	CART	**Random Forest**	Neural Network
Accuracy	77.1%	**84.2%**	65.7%
Precision Rate	75.6%	**83.2%**	74.2%
Recall	80.1%	**85.8%**	63.4%

Based on our findings, we recommend using the Random Forest model because:

1. Random Forest is overall the most accurate. It has the highest precision score, which means it can more accurately predict most fraud cases while minimizing false positives.
2. Enhanced Understanding: Our report's objective is not merely to predict occurrences of electricity theft but to also delve into the root causes driving such incidents. By identifying key contributing factors, utility companies can bolster their theft prevention strategies and mitigate future theft occurrences. As fraudsters adapt their tactics, Random Forest's flexibility allows us to capture evolving trends and pinpoint the most significant variables from a plethora of input features.
3. Non-Linear Relationship: As Random Forest combines multiple decision trees, it is effective for our dataset where the relationship between variables is not straightforward or is highly dimensional and interconnected.
4. Robustness & Handle Large Data sets: The use of bagging in Random Forest models, which involves generating diverse trees and selecting random subsets of features for node splits,

ensures the model's stability when managing large datasets. Additionally, by aggregating the predictions from multiple trees, the model achieves robustness, effectively handling variability and ensuring reliable performance.

5. SmartElec – Proposed SolutionSmartElec takes on 3 crucial processes. Firstly, identification of potential meter tampering clients. Secondly, sending employees for onsite verification and third, a dashboard to view the results and to provide managers with actionable insights.

7.1 SmartElec Phone Application

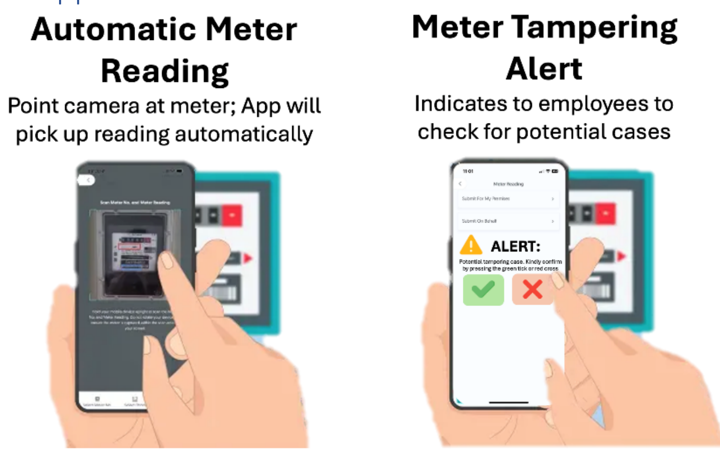


Figure 12: Phone application for frontline workers mockup

This multi-approach solution starts off with our identified random forest model, to identify frauds within the potential households, frontline employees will then conduct verification. To facilitate and enable ease of recording data, an application will be created. The key functions include the following:

1) List the households that employees are assigned to and must conduct verification.

2) In-app image recognition AI function for recording accurate readings. The purpose here is to not only enable ease of recording for employees but more crucially, it is to prevent cases of bribery where clients want to go undetected. By capturing the photo of each meter, a record can also be kept for future references. Furthermore, geolocation of where each employee is can also reinforce that they are recording the readings on site. Hence, this also addresses NTPC's pain point of employee bribery where a significant focus is placed and reiterated in their fraud prevention policy (NTPC, n.d.).

3) Confirmation if clients have tampered with their meter or not, based on physical meter check.

Moreover, these features in the application are also relevant to the new Smart Meters' meter tampering verification requirements. Hence, ensuring that the application is relevant to both before and after implementation of the Smart Meters.

7.2 SmartElec Dashboard Implementation

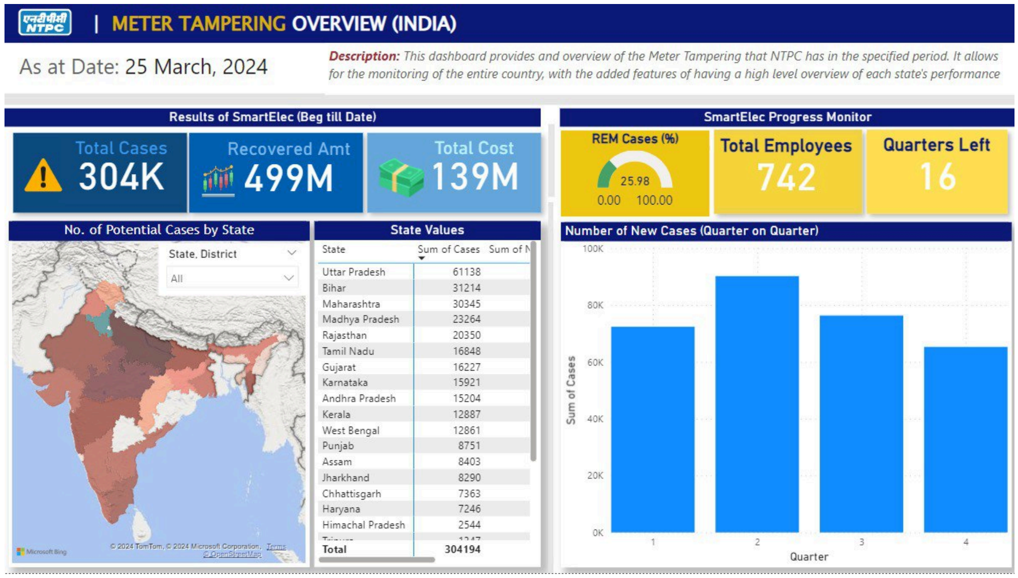


Figure 13: Dashboard Mockup

The following dashboard is designed for managers to identify key details and have a high-level overview of the SmartElec project, by state and the entire country.

Firstly, the top left portion showcases statistics related to the overall progress of the project. This encompasses data on 1) the total number of meter tampering cases checked since the start of the project, 2) recovered amount from meter tampering clients and 3) total cost incurred. Each metric is designed to provide managers with a comprehensive view of how effective the SmartElec project is at reducing meter tampering. Hence, determining if the entire project is a success or failure.

The top right portion highlights the crucial information for the completion of the project. From the left, the first yellow box indicates the percentage of cases screened for meter tampering out of all the potential cases identified by the predictive model. Hence, enabling managers to track their progress. The middle box indicates the total number of frontline employees hired to conduct checks. Lastly, the final box indicates the estimated timeline required for managers to screen all cases. These 3-information combined, delivers actional insight for managers in deciding if they are comfortable with the duration required to complete the project and if they should hire more manpower.

In addition, a heatmap has been generated to visually represent the number of cases per state, with a gradient from light to dark, low to high respectively. This visualisation can not only enable managers to view which states require more assistance but also enable NTPC to facilitate the effective deployment of their Smart Meters with the government. Despite acknowledging the susceptibility of Smart Meters to hacking, as discussed in Section 4.1, given how the Indian Ministry of Power remains steadfast in its commitment to deploying Smart Meters nationwide, this data can serve as a valuable tool for informing the prioritization of Smart Meter installations across different regions of the country, in hopes that it can help in the reduction of meter tampering cases. The information required can also be easily obtained by NTPC like, longitude and latitude of each customer's address, their district and if they have tampered with their meter or not.

Lastly, the bottom right chart has been included for managers to plan for resource management. It indicates the number of new cases detected quarterly where managers can potentially spot trends. For instance, in quarter 2, a spike has been indicated. Hence, managers can identify and research the causes. If such occurrences are seasonal, they can better manage the required manpower for the following year by hiring and training more employees in quarter 4 and 1 of the following year.

8. Evaluation, Recommendation & Future Plans

8.1 Evaluation of Smart Elec to Current Solutions

Advantage

1) Capitalise on all the benefits of Smart Meter while mitigating negative aspect of being hacked easily
2) Enhances NTPC current approach of identifying tampering cases during site visits where process is efficient and accurate (reduce human error) while preventing potential bribery.
3) Reduced complexity, prediction model enables NTPC to determine potential cases more easily compared to 2-prong approach of using decision rule-based approach.

Disadvantage

1) Resistance to new methods and technology. Existing employees who are used to the current methods or non tech savvy may be resistant to change, hindering the effectiveness of the SmartElec.
2) A limitation is that the Random Forest model like any other model is that the model can only predict and detect fraudulent activity based on the characteristics of previous fraud instances already recognized by the company. The model cannot forecast fraudulent activities that have not yet been caught by NTPC's existing anti-fraud procedures

Therefore, while cost may be a hindering factor, the mock-up illustrates a simulation of how the project will run. Moreover, as the benefits outweigh the disadvantages, in comparison to the current approach, SmartElec is still the overall better solution.

8.2 Recommendation

Firstly, for disadvantage 1, the solution would be to provide sufficient training for employees to use SmartElec and an intuitive application design. An example includes the previous application mock up with big tick and cross buttons and concise instructions.

For disadvantage 2, constraint can be mitigated when more fraud incidents are detected, and data is collected over time. These new data can be added into the prediction model, allowing it to assess a broader range of fraudulent traits. This broadens the model's scope and enables it to discover more cases that were previously missed by anti-fraud frameworks. By accurately identifying previously undetected cases, NTPC can enhance its fraud detection measures. In the long run, with better detection measures, electricity theft cases will decrease as fraudulent customers are deterred by NTPC's strong anti-fraud mechanisms.

8.3 Post-Implementation: Monitoring Impact of Recommendations on Electricity Theft

After implementing these strategies, we recommend NTPC to monitor the impact on the change in number of cases of electricity theft and its associated losses. This is because it is crucial to assess the effectiveness of the implemented measures and identify areas for further improvement.

NTPC should closely examine changes in their underwriting amounts, specifically focusing on losses attributed to electricity theft i.e. whether NTPC is continuing to suffer significant underwriting loss. In the long-run, possible impacts post-implementation include:

1. **Large reduction in electricity theft:** If the preventive measures prove to be highly effective, the company will observe a decrease in underwriting losses attributed to electricity theft. In such cases, the company should continue implementing and refining these measures to further enhance frameworks and minimize losses.
2. **Persistent theft and continued underwriting losses:** If preventive measures show limited effectiveness and underwriting losses persist, the company may need to implement additional measures. This may involve hiring experienced auditors to conduct thorough inspections and investigations to identify instances of electricity theft. Proper auditing of electricity consumption rate at the distribution level can prevent manipulation of energy meters by illegal consumers (Depuru, S., 2011). Nonetheless, the insights gained from our models can still be valuable in identifying potential fraud cases and aiding audits. By collecting more data and refining analytics models, NTPC can optimize their inspection processes and gradually reduce losses due to electricity theft.

8.4 Future Plans: Partnership with Other Utility Companies

NTPC can form partnerships with major utility companies to share electricity theft and fraud data and insights about frauds to allow the discovery of more fraudulent trends. This collaboration aims to help utility companies uncover emerging fraudulent trends more effectively. Given the widespread occurrence of electricity fraud across the industry, other major utility companies, such as Adani Electricity and Tata Power, also encounter similar challenges and might possess valuable insights into fraudulent activities in India. By sharing data, companies can gain a deeper understanding of fraudulent behaviours and identify syndicate fraudsters who engage in repeated fraudulent activities across various utility companies. Implementing a joint warning system among partnered companies can serve as a preventive measure to deter fraudsters. Additionally, access to more data also enhances the training of predictive models, resulting in a better grasp of fraudulent patterns and mutual benefits for all collaborating companies. These also prevents meter tampering clients from jumping to other companies to avoid paying their stolen electricity and stealing from another company. Given the nature of such sensitive information, all the companies can work with the government where it acts as a mediator. Furthermore, given how financially they are to the Smart Meters, there is a high likelihood that they are willing to facilitate.

9. Conclusion

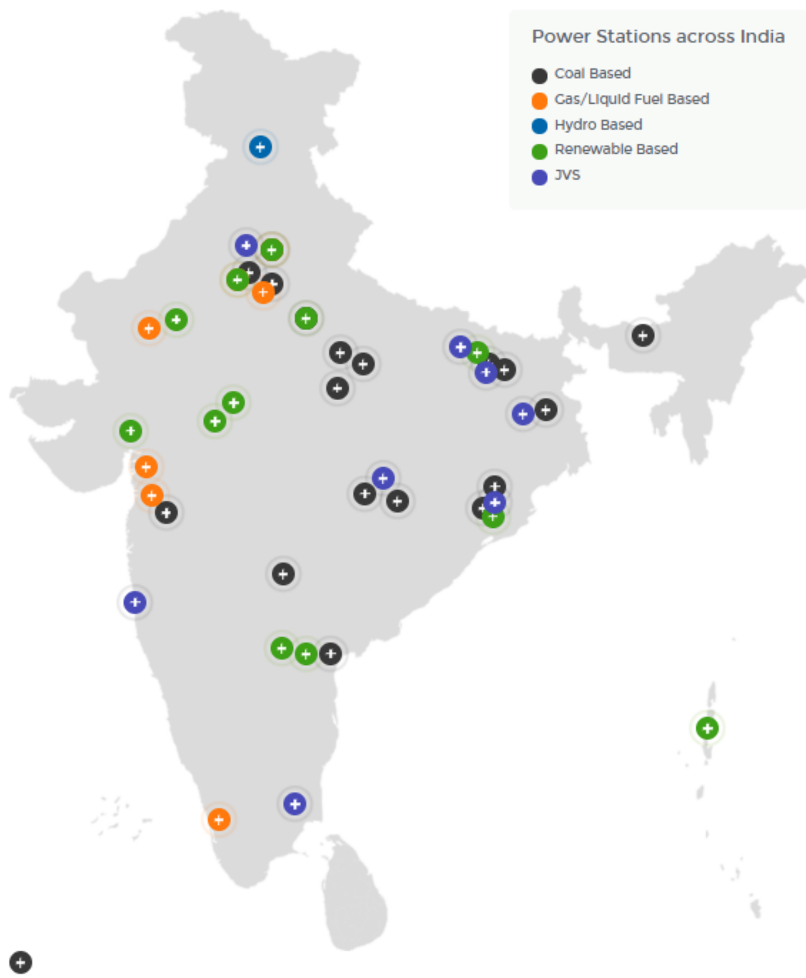
In conclusion, our chosen Random Forest machine learning model will enable NTPC to detect fraudulent electricity theft accurately and decrease underwriting losses.

Our proposed non-technical solutions can help NTPC improve its anti-fraud measures. By establishing strategic collaborations and enacting policies with both internal and external stakeholders, we can enhance NTPC's ability to detect fraud with greater precision and effectiveness.

Therefore, through the integration of machine learning models and operational enhancements, we are confident that we have met our business goal of mitigating profit loss for NTPC by identifying fraudulent customers.

Appendices

Appendix A: *NTPC Power Plants across India*



Appendix B

No.	Attribute Name	Description	Type
5	creation_date	Client's Joining Date	Continuous
6	target	Fraud or Not Fraud	Discrete (Binary)
7	invoice_date	Date of the Invoice	Continuous
8	tarif_type	Type of Tax	Discrete
9	counter_number	Counter's Number	Discrete
10	counter_code	Counter's Code	Discrete
11	reading_remarque	Notes that the STEG agent takes during inspection (e.g., If the counter shows something wrong, etc.)	Discrete
12	counter_statue	Status of the counter which has 5 values such as working fine, not working, on hold status, etc.	Discrete
13	counter_coefficient	An additional coefficient to be added when standard consumption is exceeded	Discrete
14	consommation_level_1	Consumption level 1	Continuous
15	consommation_level_2	Consumption level 2	Continuous
16	consommation_level_3	Consumption level 3	Continuous
17	consommation_level_4	Consumption level 4	Continuous
18	old_index	Old Index	Continuous
19	new_index	New Index	Continuous
20	months_number	Number of Month	Discrete
1	district	Client's District	Discrete
2	client_id	Client's Unique ID	Discrete
3	client_catg	Client's Category	Discrete
4	region	Client's Area / Region	Discrete

Variables in Dataset

Appendix C

Data Transformation for categorical variables

```
client_train.df$disrict <- factor(client_train.df$disrict)
client_train.df$client_catg <- factor(client_train.df$client_catg)
client_train.df$target <- factor(client_train.df$target)

invoice_train.df$tarif_type <- factor(invoice_train.df$tarif_type)
invoice_train.df$counter_number <- factor(invoice_train.df$counter_number)
invoice_train.df$counter_statue <- factor(invoice_train.df$counter_statue)
invoice_train.df$counter_code <- factor(invoice_train.df$counter_code)
invoice_train.df$reading_remarque <- factor(invoice_train.df$reading_remarque)
```

	client_id	invoice_date	tarif_type	counter_number	counter_statue	counter_code
1	train_Client_125864	2011-12-12	11	101545	3	413
2	train_Client_25894	2010-07-13	40	6986740	3	5
3	train_Client_3636	2012-01-18	40	312650	3	5
4	train_Client_37069	2010-10-11	11	5284	3	413
5	train_Client_44407	2008-02-28	11	66921	3	203
6	train_Client_54609	2010-04-06	11	200912	3	207
7	train_Client_66981	2006-07-17	11	132701	3	420
8	train_Client_66981	2006-07-17	11	132701	3	420
9	train_Client_7066	2014-10-28	40	4463323	3	5
10	train_Client_72519	2013-01-21	11	246464	3	203
11	train_Client_77029	2012-12-28	11	540417	3	420

Appendix D

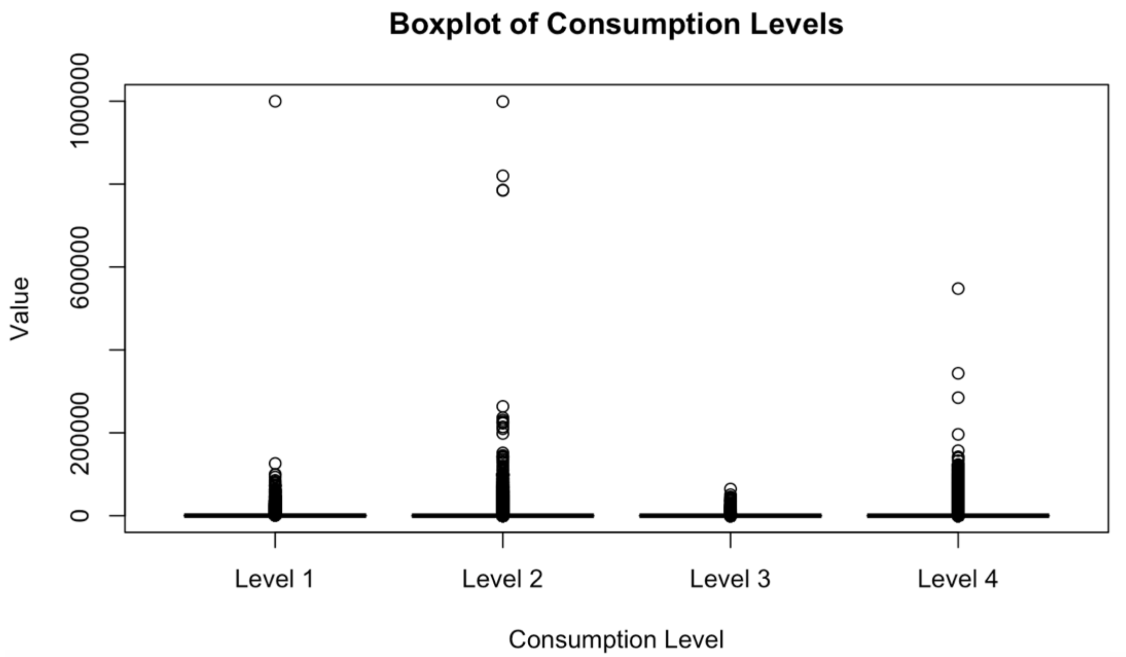
Checks for duplicate

```
#Check for duplicates
duplicated_rows <- client_train.dt[duplicated(client_train.dt), ]
View(duplicated_rows)
duplicated_rows <- invoice_train.dt[duplicated(invoice_train.dt), ]
View(duplicated_rows)

# Remove duplicate rows from client_train.dt
invoice_train.dt<- unique(invoice_train.dt)
#Final Check
duplicated_rows <- invoice_train.dt[duplicated(invoice_train.dt), ]
View(duplicated_rows)
```

Appendix E

Boxplot of the 4 consumption levels, showing the outliers for level 1, 3 & 4



Appendix F

Final Full_data_model variables

```
> full_data_model %>%
+   glimpse()
Rows: 204,902
Columns: 137
$ tarif_type                            <fct> 11, 11, 11, 11, 11, …
$ disrict                               <fct> 60, 69, 62, 69, 62, …
$ client_catg                           <fct> 11, 11, 11, 11, 11, …
$ region                                <fct> 101, 107, 301, 105, …
$ avg_consom_l_1_ELEC                   <dbl> 352.40000, 557.54054…
$ avg_consom_l_1_GAZ                    <dbl> 0.0000, 0.0000, 0.00…
$ var_consom_l_1_ELEC                   <dbl> 96313.07059, 39178.6…
$ var_consom_l_1_GAZ                    <dbl> 0.000, 0.000, 0.000,…
$ sd_consom_l_1_ELEC                    <dbl> 310.343472, 197.9359…
$ sd_consom_l_1_GAZ                     <dbl> 0.00000, 0.00000, 0.…
$ median_consom_l_1_ELEC                <dbl> 267.0, 520.0, 655.5,…
$ median_consom_l_1_GAZ                 <dbl> 0.0, 0.0, 0.0, 0.0, …
$ mode_consom_l_1_ELEC                  <dbl> 1200, 498, 407, 0, 8…
$ mode_consom_l_1_GAZ                   <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ avg_diff_consom_l_1_ELEC              <dbl> 5.735294, 16.722222,…
$ avg_diff_consom_l_1_GAZ               <dbl> 0.0000000, 0.0000000…
$ range_consom_l_1_ELEC                 <dbl> 1162, 1017, 2212, 15…
$ range_consom_l_1_GAZ                  <dbl> 0, 0, 0, 0, 0, 693, …
$ sd_cumsum_consommation_level_1_ELEC   <dbl> 3582.745636, 6080.95…
$ sd_cumsum_consommation_level_1_GAZ    <dbl> 0.0000, 0.0000, 0.00…
$ avg_cumsum_consommation_level_1_ELEC  <dbl> 7495.943, 10008.027,…
$ avg_cumsum_consommation_level_1_GAZ   <dbl> 0.000, 0.000, 0.000,…
$ median_cumsum_consommation_level_1_ELEC <dbl> 8385.0, 9538.0, 8573…
$ median_cumsum_consommation_level_1_GAZ <dbl> 0.0, 0.0, 0.0, 0.0, …
$ avg_consom_l_2_ELEC                   <dbl> 10.57143, 0.00000, 3…
$ avg_consom_l_2_GAZ                    <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ var_consom_l_2_ELEC                   <dbl> 1898.2521, 0.0000, 2…
$ var_consom_l_2_GAZ                    <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ sd_consom_l_2_ELEC                    <dbl> 43.56894, 0.00000, 1…
$ sd_consom_l_2_GAZ                     <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ median_consom_l_2_ELEC                <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ median_consom_l_2_GAZ                 <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ mode_consom_l_2_ELEC                  <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ mode_consom_l_2_GAZ                   <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ avg_diff_consom_l_2_ELEC              <dbl> 0.000000, 0.000000, …
$ avg_diff_consom_l_2_GAZ               <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ range_consom_l_2_ELEC                 <dbl> 186, 0, 682, 0, 400,…
$ range_consom_l_2_GAZ                  <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ sd_cumsum_consommation_level_2_ELEC   <dbl> 93.80803, 0.00000, 2…
$ sd_cumsum_consommation_level_2_GAZ    <dbl> 0.00000, 0.00000, 0.…
$ avg_cumsum_consommation_level_2_ELEC  <dbl> 322.2286, 0.0000, 56…
$ avg_cumsum_consommation_level_2_GAZ   <dbl> 0.00000, 0.00000, 0.…
$ median_cumsum_consommation_level_2_ELEC <dbl> 370, 0, 682, 0, 668,…
$ median_cumsum_consommation_level_2_GAZ <dbl> 0, 0, 0, 0, 0, 400, …
$ avg_consom_l_3_ELEC                   <dbl> 0.000000, 0.000000, …
$ avg_consom_l_3_GAZ                    <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ var_consom_l_3_ELEC                   <dbl> 0.00, 0.00, 0.00, 0.…
```

```
$ diff_min_new_old_index_ELEC        <dbl> 124, 551, 459, 0, 95…
$ diff_min_new_old_index_GAZ         <dbl> 0, 0, 0, 0, 0, 0, 0,…
$ max_old_index_ELEC                 <dbl> 16493, 23940, 41532,…
$ max_old_index_GAZ                  <dbl> 0, 0, 0, 0, 0, 5138,…
$ max_new_index_ELEC                 <dbl> 17078, 25022, 44614,…
$ max_new_index_GAZ                  <dbl> 0, 0, 0, 0, 0, 5518,…
$ diff_max_new_old_index_ELEC        <dbl> 585, 1082, 3082, 15,…
$ diff_max_new_old_index_GAZ         <dbl> 0, 0, 0, 0, 0, 380, …
$ sd_old_index_ELEC                  <dbl> 4527.744415, 6124.12…
$ sd_old_index_GAZ                   <dbl> 0.0000, 0.0000, 0.00…
$ sd_new_index_ELEC                  <dbl> 4579.666655, 6119.72…
$ sd_new_index_GAZ                   <dbl> 0.0000, 0.0000, 0.00…
$ diff_sd_new_old_index_ELEC         <dbl> 51.922239, -4.402326…
$ diff_sd_new_old_index_GAZ          <dbl> 0.000000, 0.000000, …
$ sd_cumsum_old_index_ELEC           <dbl> 96505.3313, 160420.1…
$ sd_cumsum_old_index_GAZ            <dbl> 0.000, 0.000, 0.000,…
$ avg_cumsum_old_index_ELEC          <dbl> 204727.857, 241770.4…
$ avg_cumsum_old_index_GAZ           <dbl> 0.0, 0.0, 0.0, 0.0, …
$ median_cumsum_old_index_ELEC       <dbl> 231834.0, 211942.0, …
$ median_cumsum_old_index_GAZ        <dbl> 0, 0, 0, 0, 0, 47807…
$ sd_cumsum_new_index_ELEC           <dbl> 100120.937582, 16643…
$ sd_cumsum_new_index_GAZ            <dbl> 0.00, 0.00, 0.00, 0.…
$ avg_cumsum_new_index_ELEC          <dbl> 212546.029, 251778.4…
$ avg_cumsum_new_index_GAZ           <dbl> 0.0, 0.0, 0.0, 0.0, …
$ median_cumsum_new_index_ELEC       <dbl> 240589.0, 221480.0, …
$ median_cumsum_new_index_GAZ        <dbl> 0.0, 0.0, 0.0, 0.0, …
$ count_counter_coefficient_ELEC     <dbl> 1, 1, 1, 1, 1, 1, 1,…
$ count_counter_coefficient_GAZ      <dbl> 0, 0, 0, 0, 0, 1, 1,…
$ count_invoice_date_ELEC            <dbl> 35, 37, 18, 20, 14, …
$ count_invoice_date_GAZ             <dbl> 0, 0, 0, 0, 0, 19, 1…
$ mode_reading_remarque_ELEC         <fct> 6, 6, 6, 6, 9, 8, 8,…
$ mode_reading_remarque_GAZ          <fct> 0, 0, 0, 0, 0, 9, 9,…
$ mode_months_number_ELEC            <dbl> 4, 4, 4, 4, 4, 4, 4,…
$ mode_months_number_GAZ             <dbl> 0, 0, 0, 0, 0, 4, 4,…
$ mode_counter_statue_ELEC           <fct> 0, 0, 0, 0, 0, 0, 0,…
$ mode_counter_statue_GAZ            <fct> 0, 0, 0, 0, 0, 0, 0,…
$ month                              <dbl> 12, 5, 3, 7, 10, 9, …
$ account_duration                   <dbl> 288, 199, 393, 269, …
$ target                             <fct> 0, 0, 0, 0, 0, 0, 0,…
```

Appendix G

CART CP for pruning model

50	0.0001967	1415	0.4285	0.871	0.00954
51	0.0001936	1431	0.4247	0.871	0.00954
52	0.0001844	1440	0.4229	0.872	0.00955
53	0.0001787	1471	0.4166	0.870	0.00954
54	0.0001770	1495	0.4110	0.870	0.00954
55	0.0001739	1527	0.4050	0.870	0.00954
56	0.0001721	1550	0.3995	0.870	0.00954
57	0.0001660	1572	0.3949	0.870	0.00954
58	0.0001581	1755	0.3630	0.872	0.00955
59	0.0001475	1762	0.3619	0.871	0.00954
60	0.0001423	1907	0.3371	0.873	0.00955
61	0.0001383	1921	0.3351	0.876	0.00957
62	0.0001352	2018	0.3203	0.877	0.00957
63	0.0001328	2056	0.3148	0.878	0.00958
64	0.0001265	2124	0.3055	0.878	0.00958
65	0.0001229	2147	0.3024	0.878	0.00958
66	0.0001180	2199	0.2948	0.818	0.00927
67	0.0001176	2215	0.2928	0.818	0.00927
68	0.0001106	2237	0.2899	0.819	0.00927
69	0.0001048	3294	0.1709	0.820	0.00928
70	0.0000948	3375	0.1618	0.823	0.00929
71	0.0000940	3409	0.1583	0.826	0.00931

Appendix H

CART Variable Importance (Unbalanced Model)

```
 [1] "region"
 [2] "avg_consom_l_1_ELEC"
 [3] "account_duration"
 [4] "var_old_index_ELEC"
 [5] "var_new_index_ELEC"
 [6] "range_old_index_ELEC"
 [7] "sd_cumsum_consommation_level_1_ELEC"
 [8] "month"
 [9] "sd_old_index_ELEC"
[10] "sd_new_index_ELEC"
[11] "median_consom_l_1_ELEC"
[12] "avg_cumsum_consommation_level_1_ELEC"
[13] "var_consom_l_1_ELEC"
[14] "sd_cumsum_old_index_ELEC"
[15] "sd_consom_l_1_ELEC"
[16] "min_new_index_ELEC"
[17] "avg_cumsum_old_index_ELEC"
[18] "avg_consom_l_1_GAZ"
[19] "median_cumsum_consommation_level_1_ELEC"
[20] "range_new_index_ELEC"
```

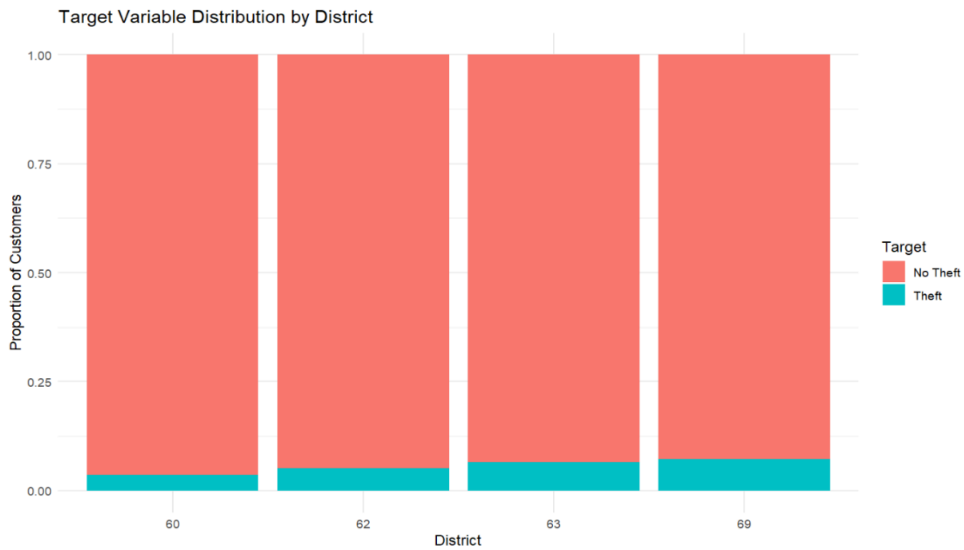
Appendix I

CART Variable Importance (Balanced Model)

```
 [1] "range_old_index_ELEC"
 [2] "var_old_index_ELEC"
 [3] "var_new_index_ELEC"
 [4] "range_new_index_ELEC"
 [5] "sd_old_index_ELEC"
 [6] "sd_new_index_ELEC"
 [7] "region"
 [8] "min_old_index_ELEC"
 [9] "min_new_index_ELEC"
[10] "avg_cumsum_old_index_ELEC"
[11] "sd_cumsum_old_index_ELEC"
[12] "median_cumsum_old_index_ELEC"
[13] "avg_consom_l_1_ELEC"
[14] "month"
[15] "account_duration"
[16] "sd_cumsum_consommation_level_1_ELEC"
[17] "avg_cumsum_consommation_level_1_ELEC"
[18] "avg_cumsum_new_index_ELEC"
[19] "median_cumsum_consommation_level_1_ELEC"
```

Appendix J

Target Distribution by District



Among the districts analyzed (60, 62, 63, and 69), district 62 exhibited the highest total number of customers, totaling 40,353, while district 63 had the lowest, with 28,987 customers. District 69 had the highest proportion of fraudulent customers, comprising 2,447 cases, accounting for 1.8% of the total fraudulent cases (5.6%). Conversely, district 60 had the lowest percentage of fraudulent cases, totaling 0.8%. In terms of honest customers, district 62 had the highest percentage, with 38,270 customers, constituting 28.2% of the total honest cases (94.4%), while district 63 had the lowest, with a percentage of 20%.

Appendix K

Target Distribution by Region

As depicted in Appendix K, region 101 stands out as the highest region for both honest and fraudulent customers, comprising 24% and 0.9% of total customers, respectively. Additionally, region 104, representing 9.5% of all customers, had 0.5% classified as fraudulent. Conversely, in region 311, fraudulent cases are higher at 0.7%, while honest customers make up 8.4%.

Furthermore, regions 199 and 206 lack any registered clients, while regions 106, 308, 379, and 399 show 0records of fraudulent customers. Overall, the data distribution indicates an imbalance.

```
Call:
 randomForest(x = features_only, y = target_variable, ntree = 100,      mt
ry = sqrt(ncol(features_only)), importance = TRUE)
               Type of random forest: classification
                     Number of trees: 100
No. of variables tried at each split: 12

        OOB estimate of  error rate: 2.71%
Confusion matrix:
     1      0  class.error
1 5247   3791 0.4194512060
0   93 134301 0.0006919952
```

Appendix L: Random Forest

Fig L1: Unbalanced Train Model Results

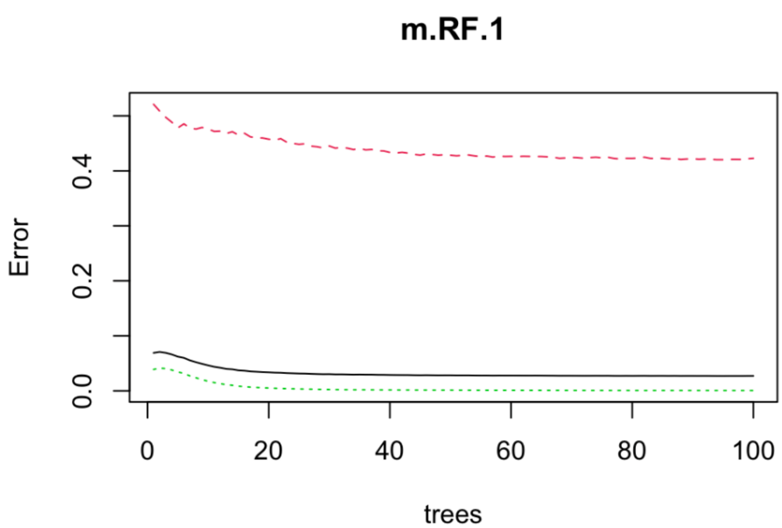


Fig L2: Unbalanced RF Plot

```
Call:
 randomForest(x = features_only, y = target_variable, ntree = 100,      mt
ry = sqrt(ncol(features_only)), importance = TRUE)
               Type of random forest: classification
                     Number of trees: 100
No. of variables tried at each split: 12

        OOB estimate of  error rate: 17.17%
Confusion matrix:
     1    0 class.error
1 7678 1376   0.1519770
0 1733 7325   0.1913226
```

Fig L3: Balanced Train Model Results

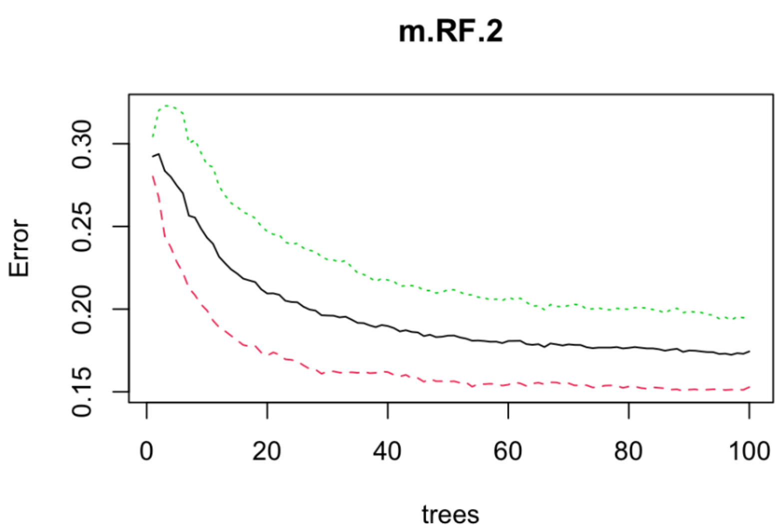


Fig L4: Balanced RF Plot

Random Forest Variable Importance

```
 [1] "region"                       "avg_consom_l_1_ELEC"
 [3] "var_consom_l_1_ELEC"          "median_consom_l_1_ELEC"
 [5] "avg_diff_consom_l_1_ELEC"     "avg_diff_old_index_ELEC"
 [7] "avg_diff_old_index_GAZ"       "avg_diff_new_index_ELEC"
 [9] "avg_diff_new_index_GAZ"       "diff_avg_new_old_index_ELEC"
[11] "diff_avg_new_old_index_GAZ"   "diff_var_new_old_index_ELEC"
[13] "min_old_index_ELEC"           "min_new_index_ELEC"
[15] "diff_min_new_old_index_ELEC" "diff_sd_new_old_index_GAZ"
[17] "month"                        "account_duration"
[19] "target"
```

Appendix N

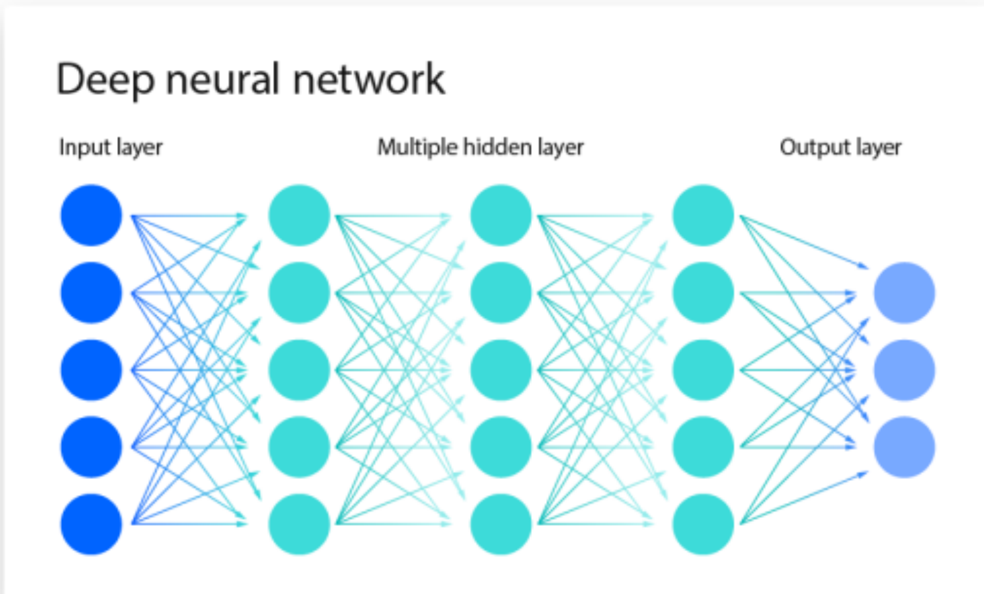
```
 [1] "disrict"
 [2] "region"
 [3] "avg_consom_l_1_ELEC"
 [4] "avg_consom_l_1_GAZ"
 [5] "var_consom_l_1_GAZ"
 [6] "median_consom_l_1_ELEC"
 [7] "mode_consom_l_1_ELEC"
 [8] "mode_consom_l_1_GAZ"
 [9] "avg_diff_consom_l_1_ELEC"
[10] "sd_cumsum_consommation_level_1_ELEC"
[11] "avg_diff_consom_l_2_ELEC"
[12] "median_cumsum_consommation_level_2_ELEC"
[13] "avg_diff_old_index_ELEC"
[14] "avg_diff_old_index_GAZ"
[15] "avg_diff_new_index_ELEC"
[16] "avg_diff_new_index_GAZ"
[17] "diff_avg_new_old_index_ELEC"
[18] "diff_avg_new_old_index_GAZ"
[19] "diff_var_new_old_index_ELEC"
[20] "diff_var_new_old_index_GAZ"
[21] "min_old_index_ELEC"
[22] "min_new_index_ELEC"
[23] "min_new_index_GAZ"
[24] "diff_min_new_old_index_ELEC"
[25] "diff_min_new_old_index_GAZ"
[26] "max_new_index_ELEC"
[27] "diff_max_new_old_index_ELEC"
[28] "diff_max_new_old_index_GAZ"
[29] "diff_sd_new_old_index_ELEC"
[30] "diff_sd_new_old_index_GAZ"
[31] "count_invoice_date_ELEC"
[32] "count_invoice_date_GAZ"
[33] "month"
[34] "account_duration"
```

```
 [1] "region"
 [2] "avg_consom_l_1_ELEC"
 [3] "var_consom_l_1_ELEC"
 [4] "median_consom_l_1_ELEC"
 [5] "avg_diff_consom_l_1_ELEC"
 [6] "avg_diff_old_index_ELEC"
 [7] "avg_diff_old_index_GAZ"
 [8] "avg_diff_new_index_ELEC"
 [9] "avg_diff_new_index_GAZ"
[10] "diff_avg_new_old_index_ELEC"
[11] "diff_avg_new_old_index_GAZ"
[12] "diff_var_new_old_index_ELEC"
[13] "min_old_index_ELEC"
[14] "min_new_index_ELEC"
[15] "diff_min_new_old_index_ELEC"
[16] "diff_sd_new_old_index_GAZ"
[17] "month"
[18] "account_duration"
```

Random Forest Comparison

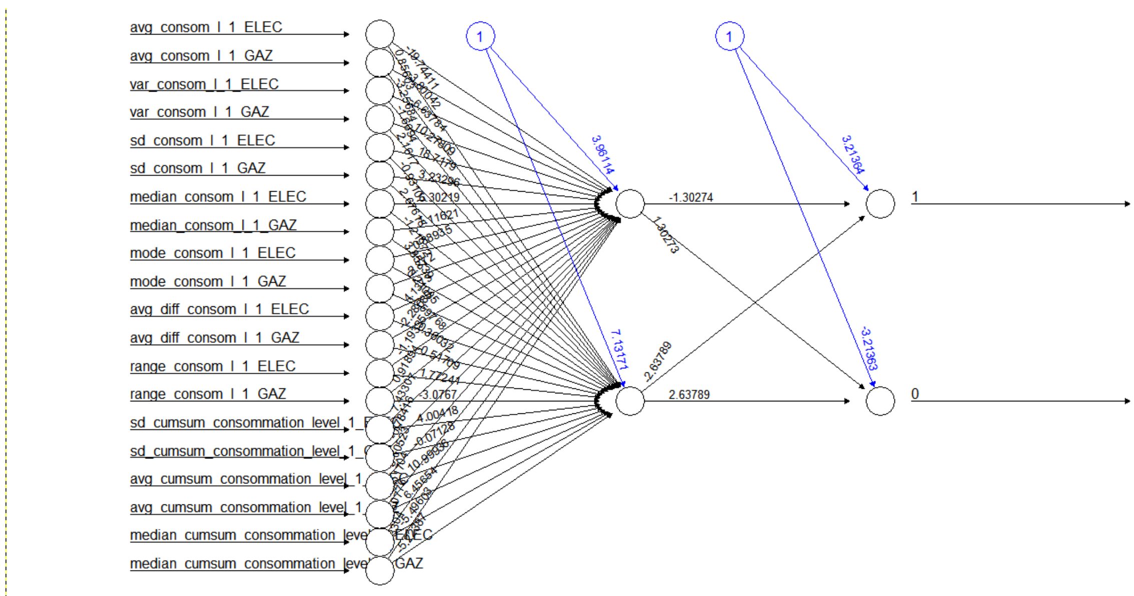
Appendix M

Neural Network Model Structure Overview



Appendix N

Neural Network Architecture

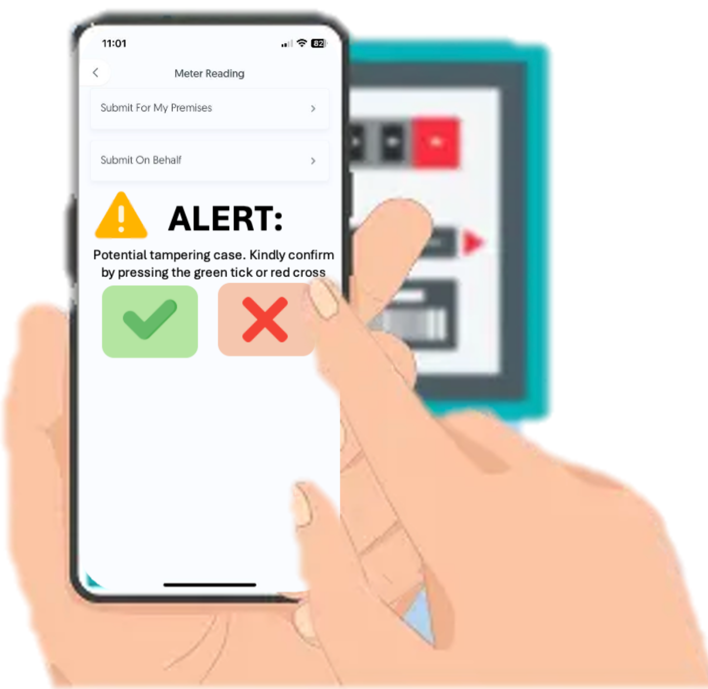


Appendix P

Intuitive Application Design Mockup

Intuitive Features

Big buttons & clear concise instructions



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